

Requirement traceability (ISRO PS-26169)

Every row maps to the module that implements it and the automated test that proves it (npm test; spec tests are in tests/spec.test.mjs).

#	Requirement	Implementation	Test / evidence
1	Screen $\geq 2000 \times 2000$, optional user-defined	config.js screenSize (clamped ≥ 2000), GUI	spec #1 (3000 px screen tracked)
2	Monochrome focal-plane array; colour optional	render.js: mono single-channel frame (default); cameraType: colour renders 3 independently-noisy RGB channels, tracker uses luminance	spec #2 (both modes)
3	Resolution 640×480 , optional user-defined	camW, camH	spec #3 (800×600)
4	FOV default $4^\circ \times 3^\circ$, user-defined	fovX, camScale()	spec #4 (FOV 2° and 8°)
5	Camera update ≥ 30 Hz	fixed step dt = 1/fps, fps ≥ 30	spec #5
6	Initial camera at screen centre	gimbal.js	spec #6
7	Target type: beacon spot	render.js addBeacon	spec #7-8
8	1 target mandatory, multiple optional	numTargets (decoys)	spec #7-8, decoy test
9	Shape user-defined, default square	square / circle / diamond	spec #9
10	Size 5–20 px, default 10	targetSize	spec #10
11	Initial location user-defined, default random	startMode, startX/Y	spec #11
12	≥ 4 motions; optional spiral, sinusoidal, user-defined	motion.js (7 motions)	spec #12 (all 7 tracked, constant speed verified in engine tests)
13-14	Max pan/tilt 5–10 $^\circ$ /s, default 5	gimbal.js	spec #13-14 (per-frame move never exceeds limit at 5 and 10 $^\circ$ /s)
15	Update interval ≥ 20 Hz	control computed every frame	spec #15
16	Acquisition ≤ 2 s	overview acquisition, tracker.js	spec #16-20, bench A/B (0.6–1.9 s)

#	Requirement	Implementation	Test / evidence
17	Tracking error ≤ 10 px	IMM + controller	spec #16-20, bench (0.1–6.7 px; C1 fails: 19 px)
18	Target loss < 5 %	state machine	spec #16-20, bench
19	Re-acquisition ≤ 1 s	COAST + overview assist	spec #19 (camera knocked off; 0.5 s dropouts)
20	Processing ≥ 20 FPS	ROI detection, 2–8 ms/frame	spec #16-20
21.1	Salt & pepper ~ 10 %, Gaussian, Poisson, selectable	render.js addNoise	spec #21.1
21.2	Noise σ up to 20	gaussianSigma	spec #21.2 (measured σ)
21.3	Jitter ± 20 px/frame	simulation.js _jitter	spec #21.3 (bound verified)
21.4	Clear, haze, fog, rain, low light, user-defined reduction	ATMOSPHERES, severity	spec #21.4 (contrast drops, still tracked)
21.5	Platform ± 20 px/frame; linear default; circular, random, spiral, figure-8	makePlatformMotion	spec #21.5 (step bound + tracked for all 5)
—	Turbulence / vibration	render.js, simulation.js	spec turbulence test
—	Real-time statistics display	App.jsx tiles, plot, minimap	browser run (screenshot)
—	Performance log (duration, FPS, acquisition, avg/max error, lock retention, processing time)	metrics.js JSON/CSV/HTML, auto-save at end of run	spec "performance log" test
—	Benchmark 1: scenarios + centroiding-error log + auto logs	scenario suite, scripts/bench.mjs	RESULTS.md
—	Benchmark 2: .mp4 @30 fps, PTZ bypassed, centroid error, RMSE, acq/re-acq, lock, FPS	videoTracker.js, videoRunner.js	npm run e2e (H.264 video, ground truth, RMSE 0.07 px)
—	Standalone executable	Electron shell (npm run build:exe) and single offline HTML (npm run build:single)	launched on Linux
—	Source code modular and documented	src/engine/*	README, this report

#	Requirement	Implementation	Test / evidence
—	Technical report, user manual	docs/TECHNICAL_REPORT.md, docs/USER_MANUAL.md	—